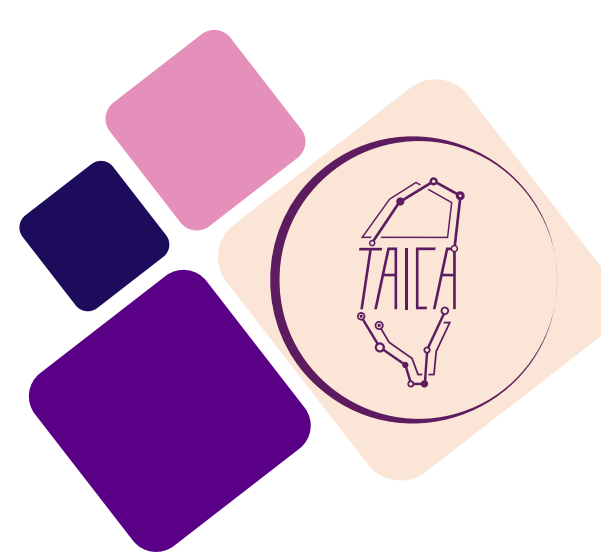




Introduction

- This task focuses on **predicting user preferences in head-to-head comparisons of chatbot responses**.
- Using a dataset from the Chatbot Arena, featuring conversations generated by various large language models (LLMs).
- The goal is to create an effective machine learning model to enhance chatbot interactions with users.
- The outcome aims to ensure chatbots align more closely with human preferences.

Data Preprocessing

- Split**
Training Set: 51,729 rows
Validation Set: 5,748 rows

- Tokenization (TinyBERT Tokenizer)**
Transform input strings into a dictionary that contains two components: `token\_ids` and `padding\_mask`.

Data Augmentation

- Method -- Random Ordering in Collation Function

```
if (random.rand() > 0.5): # 50% with original ordering
    for item in batch:
        trunc_prompt = random_truncate_prompt(item['prompt'])
        all_prompt.append(trunc_prompt)

        trunc_a = random_truncate_prompt(item['response_a'])
        all_prompt_response_a.append(trunc_a)
        trunc_b = random_truncate_prompt(item['response_b'])
        all_prompt_response_b.append(trunc_b)

        all_id.append(item['id'])
        if (item['winner_model_a'] == 1):
            all_winner.append(0) # class 0: response_a winner
        elif (item['winner_model_b'] == 1):
            all_winner.append(1) # class 1: response_b winner
        else:
            all_winner.append(2) # class 2: tie
else: # 50% flip the ordering and label of the a, b response
    for item in batch:
        trunc_prompt = random_truncate_prompt(item['prompt'])
        all_prompt.append(trunc_prompt)

        trunc_a = random_truncate_prompt(item['response_b'])
        all_prompt_response_a.append(trunc_a)
        trunc_b = random_truncate_prompt(item['response_a'])
        all_prompt_response_b.append(trunc_b)

        all_id.append(item['id'])
        if (item['winner_model_a'] == 1):
            all_winner.append(1) # flip label
        elif (item['winner_model_b'] == 1):
            all_winner.append(0) # flip label
        else:
            all_winner.append(2) # class 2: tie
```

Operation: 50% chance of **swapping the order** of `response\_a` and `response\_b`, as well as flipping their corresponding labels.

Labels:
If `winner\_model\_a` equals 1, then the label is 0, indicating that `response\_a` is better.
If `winner\_model\_b` equals 1, then the label is 1, indicating that `response\_b` is better.
If neither option is selected, label it as 2 (indicating a `tie`).

- Method -- Random Truncation Text Function

```
def random_truncate_prompt(prompt, max_length=max_token):
    """
    Randomly truncates a prompt before tokenization.

    """
    words = prompt.split() # Split by whitespace
    num_words = len(words)

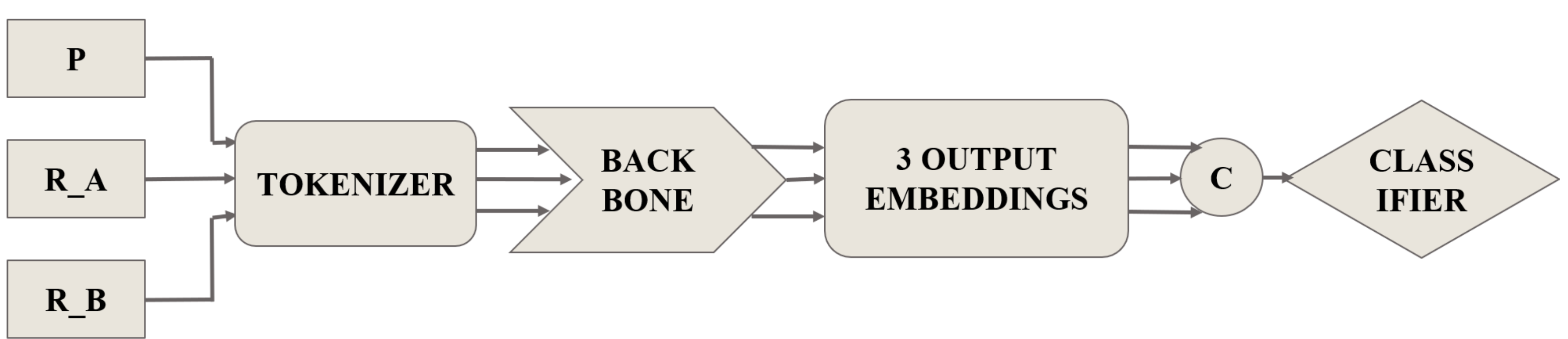
    if num_words > max_length:
        truncatable_length = num_words - max_length
        start_idx = random.randint(0, truncatable_length)
        truncated_prompt = " ".join(words[start_idx:start_idx + max_length])
    else:
        truncated_prompt = prompt

    return truncated_prompt
```

Operation: Randomly truncates the text by:

- Step 1.** Divide the text into individual words.
- Step 2.** Ensure that **the text does not exceed the maximum length**.
- Step 3.** **Randomly select a contiguous subset of words**, ensuring that it does not exceed a specified maximum length if the text is excessively long.

Framework



Training Configurations

Batch Size	Learning Rate	Epochs	Weight Decay	Max Length	Pre-Trained Model	Optimizer	Loss Function
16	0.00003	2	0.01	512	TinyBERT	AdamW	Log Loss

Experiments

Experiment	Method	Model	Data Augmentation	Validation Accuracy	Validation F1 Score
Case 1	Baseline	TinyBERT	No	0.4548	0.4346
Case 2	Our	TinyBERT	No	0.4763	0.4644
Case 3	Our	TinyBERT	Yes	0.4828	0.4711

Conclusion

We have **adjusted the selection of pre-trained model** based on the baseline method, **modified the input structure** of the model, and **applied data augmentation during preprocessing**. The numerical results indicate that these methods effectively enhance the model's performance, resulting in higher accuracy and F1 score.